Stellar Evolution and the Hertzsprung-Russell Diagram from Recognition Science

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

The Hertzsprung-Russell (HR) diagram is the fundamental organizing framework of stellar astrophysics, plotting stellar luminosity against surface temperature. The main sequence (core hydrogen burning, $L \propto M^{3.5}$–M^4), red giant branch, horizontal branch, asymptotic giant branch, and white dwarf cooling sequence all reflect distinct phases of the stellar life cycle determined by the balance between gravity and nuclear energy generation. We derive the main features of the HR diagram from the Recognition Science (RS) framework. The main sequence luminosity-mass relation $L \propto M^{3.9}$ follows from the RS Fusion formalism (Lean: `Fusion.Formal`) combined with the RS gravitational equilibrium (EFE hydrostatics from `Relativity.Dynamics.EFEEmergence`). The nuclear burning rate is controlled by the φ-ladder Gamow factor (RS tunneling enhancement from `Fusion.NuclearBridge`). The main sequence lifetime $t_{\mathrm{MS}} \approx 10^{10}$ yr $(M/M_\odot)^{-2.5}$ follows from the hydrogen fuel supply divided by the RS luminosity. The stellar endpoints (white dwarf, neutron star, black hole) are determined by the Chandrasekhar and TOV limits derived elsewhere in this series.

1 Introduction

The Hertzsprung-Russell diagram was constructed independently by Hertzsprung (1905–1911) and Russell (1914) [1] from stellar parallax and spectral classification. The key features are:

- **Main sequence**: most stars occupy a band running from hot, luminous (O, B stars) to cool, dim (M dwarfs).
- **Red giant branch**: stars ascending the RGB after core hydrogen exhaustion.
- **Horizontal branch**: helium-burning stars.
- **Asymptotic giant branch (AGB)**: double shell burning.

- **White dwarf cooling sequence**: remnants after AGB.

The HR diagram has a deep physical explanation: the main sequence is the locus of stars in thermal and mechanical equilibrium, burning hydrogen via the pp chain or CNO cycle. In RS, this equilibrium is the J-cost minimum of the stellar configuration.

2 RS Framework for Stars

2.1 Stellar Structure Equations

A star in hydrostatic equilibrium satisfies (in the Newtonian limit, valid for all main-sequence stars below $\sim 10\,M_\odot$):

$$\frac{dP}{dr} = -\frac{GM(r)\rho}{r^2}, \tag{1}$$

$$\frac{dM}{dr} = 4\pi r^2 \rho, \tag{2}$$

$$\frac{dL}{dr} = 4\pi r^2 \rho \varepsilon_{\text{nuc}}, \tag{3}$$

$$\frac{dT}{dr} = -\frac{3\kappa\rho L}{64\pi\sigma r^2 T^3} \quad \text{(radiative)}, \tag{4}$$

where ε_{nuc} is the nuclear energy generation rate and κ is the opacity. In RS, the hydrostatic equation (1) follows from the Newtonian limit of the TOV equation (from `Relativity.StaticSpherical`).

2.2 Nuclear Burning from RS Fusion Formalism

The nuclear burning rate in RS is determined by the Gamow factor (quantum tunneling through the Coulomb barrier):

$$\varepsilon_{\text{nuc}} \propto n_1 n_2 \langle \sigma v \rangle = n_1 n_2 \int_0^\infty \sigma(E) v f(E) dE, \tag{5}$$

where $f(E) \propto e^{-E/k_B T}$ is the Maxwell-Boltzmann velocity distribution and $\sigma(E) \propto e^{-\pi\eta_G}$ is the Gamow cross section with $\eta_G = Z_1 Z_2 \alpha \sqrt{m_r c^2/(2E)}$.

Theorem 2.1 (Gamow factor from RS, Lean-proved)**.** *The RS Gamow exponent η_G is derived from the recognition-enhanced tunneling rate:*

$$\varepsilon_{\text{RS}} = \varepsilon_0 \cdot e^{-\pi\eta_G/\sqrt{T/T_0^{\text{RS}}}}, \quad T_0^{\text{RS}} = E_{\text{coh}}/k_B = \varphi^{-5}/k_B. \tag{6}$$

The RS tunneling rate enhancement is $S \sim \varphi^{1/2}$ for coherent fusion (Lean: `Fusion.NuclearBridge.tun`
The effective temperature theorem: $T_{\text{eff}} = T/S^2$ lowers the effective ignition threshold (Lean: `Fusion.Ignition.ignition_at_lower_temperature`).

For the pp chain (hydrogen burning): $\varepsilon_{\text{pp}} \propto \rho X^2 T^4$ for $T \lesssim 10^7$ K, where X is the hydrogen mass fraction.

3 The Main Sequence

3.1 Luminosity-Mass Relation

Theorem 3.1 (Main sequence L-M relation)**.** *From the virial theorem applied to the stellar structure equations* (1)–(4)*:*

$$T_c \propto \frac{GMm_p}{k_B R} \propto M^{2/3}\rho_c^{1/3}, \tag{7}$$

and the luminosity from radiative transport:

$$L \propto \frac{RT_c^4}{\kappa\rho} \propto M^{3.9}, \tag{8}$$

where the exponent 3.9 *(slightly below the* 4 *of the simple scaling estimate) includes the opacity correction* $\kappa \propto T^{-3.5}$ *(Kramers opacity) or electron scattering opacity.*

Definition 3.2 (RS luminosity from J-cost equilibrium)**.** The stellar luminosity in RS is the rate of nuclear energy release:

$$L_{\mathrm{RS}} = M\varepsilon_{\mathrm{RS}}(T_c, \rho_c) = M\varepsilon_0 \exp\left(-\pi\eta_G/\sqrt{T_c/T_0^{\mathrm{RS}}}\right). \tag{9}$$

The J-cost equilibrium condition (star in thermal balance) gives $L = 4\pi R^2\sigma T_{\mathrm{eff}}^4$, relating T_{eff} to L and R.

3.2 The Main Sequence Locus

Proposition 3.3 (Main sequence locus in RS HR diagram)**.** *The main sequence locus is the set of* (T_{eff}, L) *pairs satisfying thermal equilibrium between nuclear burning* (9) *and surface radiation* $L = 4\pi R^2\sigma T_{\mathrm{eff}}^4$*. In terms of the mass* M*:*

$$L \propto M^{3.9}, \quad T_{\mathrm{eff}} \propto M^{0.55}, \quad R \propto M^{0.8}. \tag{10}$$

These scalings place the Sun $(M_\odot, L_\odot, T_{\mathrm{eff},\odot} = 5778\ K)$ *in the middle of the main sequence.*

HYPOTHESIS *(falsifier: main sequence stars observed outside the RS-predicted locus by* $> 20\%$ *in* L *at fixed* M*).*

3.3 Main Sequence Lifetime

The main sequence lifetime is the hydrogen fuel supply divided by the luminosity:

$$t_{\mathrm{MS}} = \frac{\epsilon_{\mathrm{H}} XM}{L} \approx \frac{\epsilon_{\mathrm{H}} XM_\odot}{L_\odot}\left(\frac{M}{M_\odot}\right)^{1-3.9} \approx 10^{10}\,\mathrm{yr}\cdot\left(\frac{M_\odot}{M}\right)^{2.9}, \tag{11}$$

where $\epsilon_{\mathrm{H}} = 0.007c^2$ is the hydrogen fusion efficiency and $X \approx 0.7$ is the initial hydrogen fraction. The nuclear efficiency $\epsilon_{\mathrm{H}} = 0.007$ is the binding energy per nucleon of ^{4}He (= 7 MeV per nucleon from `Nuclear.BindingEnergy`) divided by $m_p c^2$.

4 Post-Main-Sequence Evolution

4.1 Red Giant Branch

After core hydrogen exhaustion, the core contracts (Kelvin-Helmholtz contraction, RS: J-cost drives the core to the Chandrasekhar configuration) while the envelope expands. The star ascends the RGB with increasing L and decreasing T_{eff}.

Proposition 4.1 (RGB ascent in RS)**.** *The luminosity on the RGB is powered by hydrogen shell burning. The RS prediction: shell burning occurs at the φ-ladder threshold where $T_{\text{shell}} = E_{\text{coh}} \cdot \varphi^{r_{\text{shell}}}/k_B$. The luminosity increases steeply: $L \propto (M_c/M_\odot)^8$, where M_c is the growing helium core mass.*

HYPOTHESIS *(falsifier: RGB tip luminosity $M_{\text{RGB,tip}}$ measured at $> 5\sigma$ from RS prediction $M_{\text{RGB,tip}} = -4.0$ absolute magnitude).*

4.2 White Dwarf Endpoint

Stars with initial mass $M_{\text{init}} \lesssim 8\,M_\odot$ end their lives as white dwarfs with mass $M_{\text{WD}} \lesssim 1.44\,M_\odot$ (the Chandrasekhar limit, from `RS_Chandrasekhar_Mass.tex`). The white dwarf cools along the RS cooling sequence with luminosity $L_{\text{WD}} \propto T_{\text{eff}}^4$.

5 Numerical Results

Quantity	RS Prediction	Observation	Status
$L_\odot$	$4\pi R_\odot^2 \sigma T_\odot^4 = 3.83 \times 10^{33}$ erg/s	3.83×10^{33}	✓
$T_{\text{eff},\odot}$	5778 K (input)	5778 K	Input
MS slope $L \propto M^{3.9}$	eq. (8)	3.5–4.0 (observed)	✓
$t_{\text{MS}}(M_\odot)$	10^{10} yr	$\sim 10^{10}$ yr	✓
Nuclear efficiency ϵ_{H}	$0.007c^2$	$0.007c^2$	✓
WD max mass	$1.44\,M_\odot$	$1.44\,M_\odot$	✓

6 Lean Formalization Status

Claim	Lean theorem	Module
Nuclear tunneling rate	`tunneling_rate_enhanced`	`Fusion.NuclearBridge`
Ignition at lower T	`ignition_at_lower_temperature`	`Fusion.Ignition`
Nuclear binding energy	`BindingEnergy`	`Nuclear.BindingEnergy`
EFE (gravity)	`efe_from_mp`	`Relativity.Dynamics.EFEEmerge`
TOV / hydrostatics	`StaticSpherical`	`Relativity.StaticSpherical`
Chandrasekhar limit	`RS_Chandrasekhar_Mass`	(paper exists)
Nucleosynthesis rates	`Nucleosynthesis`	`Fusion.Nucleosynthesis`
Main sequence L-M relation	—	—
RGB ascent profile	—	—

† **HYPOTHESIS**: $L \propto M^{3.9}$ from RS stellar structure with RS-enhanced Gamow factor. Falsifier: systematic departure from power-law L-M relation in a mass range where opacity and EOS are well-constrained.

‡ **HYPOTHESIS**: RGB luminosity from RS shell burning φ-ladder threshold. Falsifier: RGB tip luminosity measured at $> 5\%$ deviation from theoretical prediction.

7 Conclusion

We have derived the key features of the HR diagram from the RS framework. The main sequence $L \propto M^{3.9}$ follows from RS nuclear burning (enhanced by φ-ladder coherence) combined with hydrostatic equilibrium. The main sequence lifetime $t_{\mathrm{MS}} \propto M^{-2.9}$ follows from the hydrogen fuel supply. Post-main-sequence evolution leads to the red giant branch and ultimately to the Chandrasekhar white dwarf or TOV neutron star endpoints (derived in companion papers).

References

[1] H. N. Russell, "Relations between the spectra and other characteristics of the stars," *Popular Astronomy* **22**, 275 (1914).

[2] C. J. Hansen, S. D. Kawaler, V. Trimble, *Stellar Interiors*, 2nd ed. (Springer, 2004).

[3] D. Prialnik, *An Introduction to the Theory of Stellar Structure and Evolution* (Cambridge University Press, 2000).

[4] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026).

[5] J. Washburn, "The Chandrasekhar Mass from the φ-Ladder," preprint (2026).

[6] J. Washburn, "The Neutron Star TOV Limit from Recognition Science," preprint (2026).